

# Unifying Random Fourier Features and Leverage Scores for Kernel Matrix Approximation

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Christopher Musco, Massachusetts Institute of Technology

ICML 2017:

“Random Fourier Features for Kernel Ridge Regression:  
Approximation Bounds and Statistical Guarantees”

Joint work with:

Haim Avron (TAU)  
Michael Kapralov (EPFL)  
Cameron Musco (MIT)  
Ameya Velingker (EPFL)  
Amir Zandieh (EPFL)

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Study **Fourier kernel approximation methods** from a **matrix sampling point of view**.

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**Lots of open questions and directions for future work.**

Opportunities to combine techniques from randomized linear algebra and Fourier methods.

Quick refresher on kernel methods.

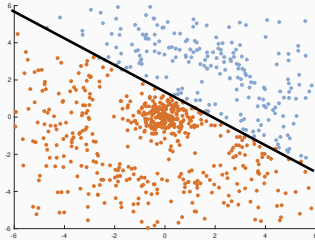
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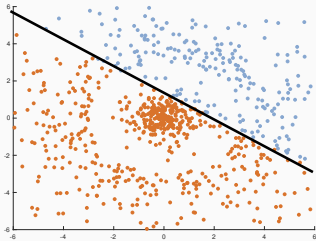


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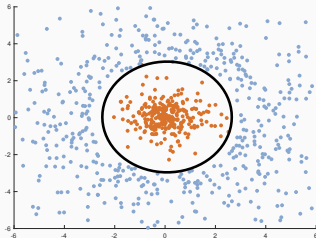
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Kernel Classifier



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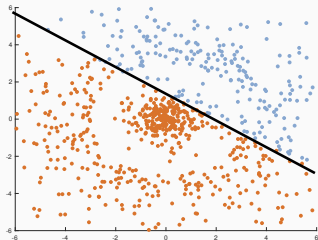
“Lift” data points to a higher dimensional feature space. E.g.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \implies \phi(\mathbf{x}) = \begin{bmatrix} x_1 \\ \vdots \\ x_d \\ x_1x_1 \\ x_1x_2 \\ \vdots \\ x_dx_d \end{bmatrix}$$

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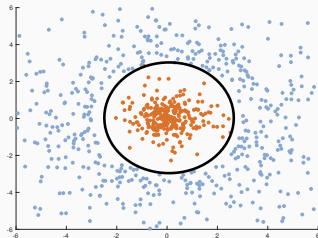
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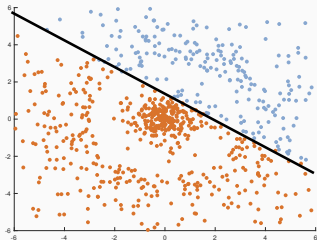


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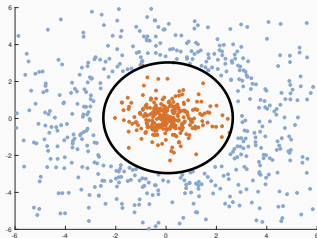
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$$x_1^2 + x_2^2 \geq 10$$
$$\phi(\mathbf{x})_3 + \phi(\mathbf{x})_4 \geq 10$$

**Main computational issue:** Forming  $\phi(\mathbf{x})$  is intractable even for moderately complex kernels.

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Can often be computed much more quickly than  $\phi(\mathbf{x}), \phi(\mathbf{y})$ .

## THE “KERNEL TRICK”

Kernel dot product can often be computed **implicitly** without forming  $\phi(\mathbf{x})$  and  $\phi(\mathbf{y})$ . For example:

$$\begin{aligned}(1 + \langle \mathbf{x}, \mathbf{y} \rangle)^2 &= (1 + x_1 y_1 + \dots + x_d y_d)^2 \\&= (1 + x_1 y_1 + x_1^2 y_1^2 + 2x_1 y_1 x_2 y_2 + \dots) \\&= \left\langle [1, x_1, x_1^2, \sqrt{2}x_1 x_2, \dots], [1, y_1, y_1^2, \sqrt{2}y_1 y_2, \dots] \right\rangle \\&= \langle \phi(\mathbf{x}), \phi(\mathbf{y}) \rangle\end{aligned}$$

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The corresponding feature map is sometimes less clear than our simple polynomial example.

The **kernel function**  $k(\mathbf{x}, \mathbf{y}) = (1 + \mathbf{x}^T \mathbf{y})^2$  provides an alternative **similarity metric** to the standard dot product.

# THE “KERNEL TRICK”

Kernel learning pipeline for data points  $\mathbf{x}_1, \dots, \mathbf{x}_n$ :

1. Choose kernel function  $k(\mathbf{x}_i, \mathbf{x}_j)$ :

$$(1 + \mathbf{x}_i^T \mathbf{x}_j)^q, e^{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}, e^{-\|\mathbf{x}_i - \mathbf{x}_j\|_1}, \text{ etc.}$$

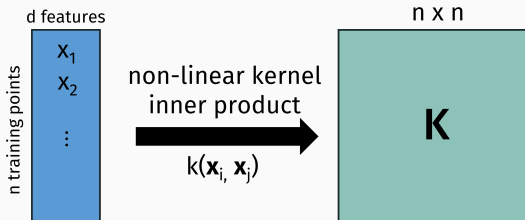
2. Form  $n \times n$  **kernel matrix**  $\mathbf{K}$  with:

$$\mathbf{K}_{i,j} = k(\mathbf{x}_i, \mathbf{x}_j)$$

3. Compute model using  $\mathbf{K}$ : compute  $(\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{b}$  for kernel regression, eigendecomposition of  $\mathbf{K}$  for kernel PCA, etc.

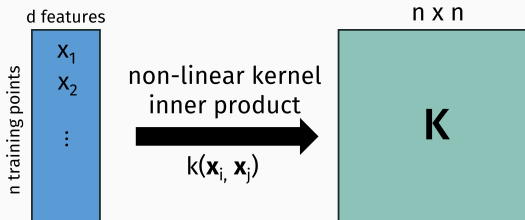
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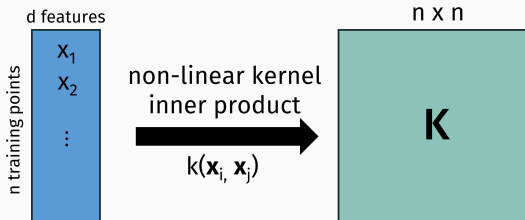
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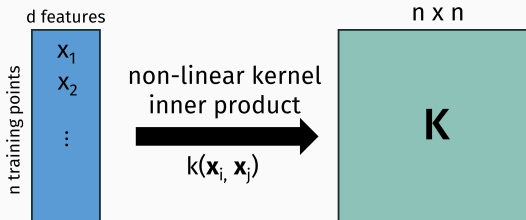
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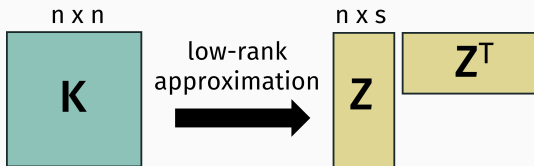


- $n = 100,000 \Rightarrow 10$  billion entries  $\Rightarrow 80$  GB to store **K**.
- Just writing down **K** requires  $\Omega(n^2)$  time.
- Other operations require even more. A single iteration for a linear system solver takes  $\Omega(n^2)$  time.

New algorithmic ideas are needed to scale kernel methods.  
Even for moderately large datasets.

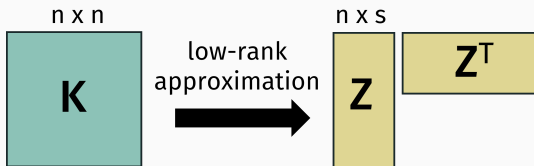
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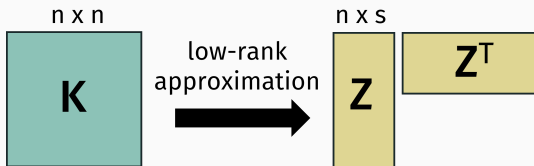
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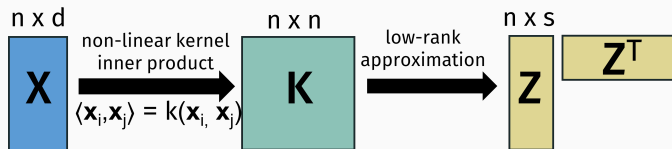


We can typically set  $s \ll n$ .

- $\mathbf{Z}$  takes  $O(ns)$  space to store.
- Orthogonalization, eigendecomposition, and inversion of  $\mathbf{Z}\mathbf{Z}^T$  all take just  $O(ns^2)$  time.
- $\mathbf{Z}\mathbf{Z}^T\mathbf{x}$  can be computed in  $O(ns)$  time.

# IMPLICIT LOW-RANK APPROXIMATION

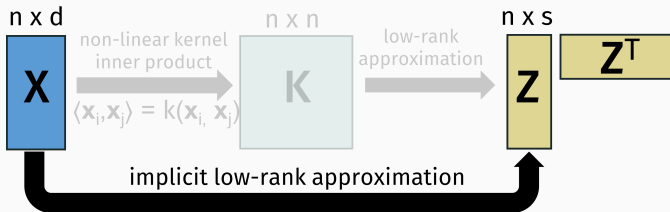
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Write  $k(\mathbf{\Delta})$  using its inverse Fourier transform,

$$p(\boldsymbol{\eta}) = \mathcal{F}^{-1}(k) \qquad k(\mathbf{\Delta}) = \int_{\boldsymbol{\eta} \in \mathbb{R}^d} p(\boldsymbol{\eta}) e^{-i\pi \boldsymbol{\eta}^\top \mathbf{\Delta}} d\boldsymbol{\eta}$$

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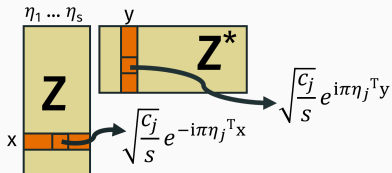
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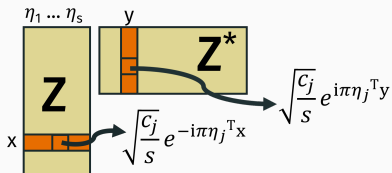
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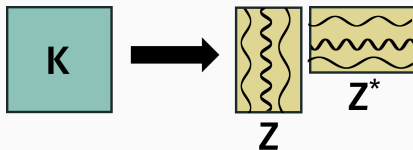
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$$\begin{aligned} Z(\mathbf{x})Z(\mathbf{y})^* &= \sum_j \frac{c_j}{s} e^{-i\pi \eta_j^T (\mathbf{x} - \mathbf{y})} \\ &\approx k(\mathbf{x} - \mathbf{y}) \end{aligned}$$

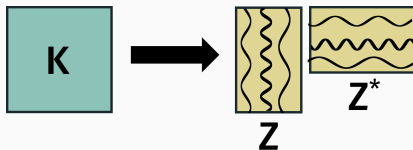
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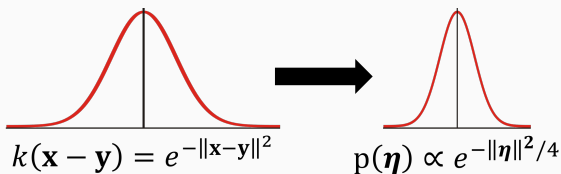
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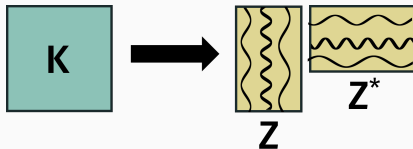
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Example (Gaussian kernel):

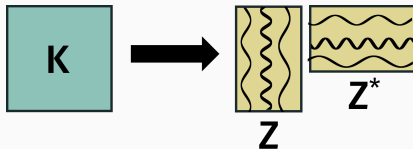


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Sample frequencies from distribution  $p$ . Set  $c_j = 1$  for all  $j$ .

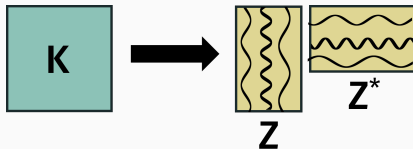
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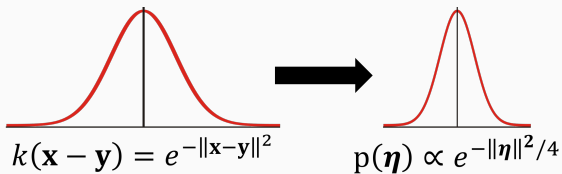
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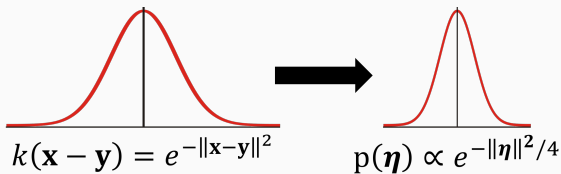
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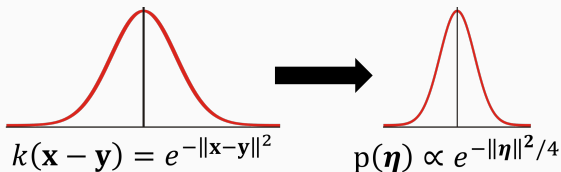
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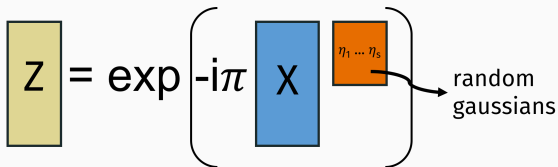
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- By Chernoff bound, if we take  $O\left(\frac{\log n}{\epsilon^2}\right)$  samples, we approximate every entry in  $\mathbf{K}$  to error  $\pm\epsilon$ .

# RANDOM FOURIER FEATURES (RFF) ALGORITHM

Super simple algorithm. For Gaussian kernel  
 $k(\mathbf{x} - \mathbf{y}) = e^{-\|\mathbf{x} - \mathbf{y}\|_2^2}$ :



The diagram illustrates the RFF algorithm equation:  $Z = \exp\left(-i\pi X G\right)$ . The variable  $Z$  is represented by a yellow rectangle. The variable  $X$  is represented by a blue rectangle. The variable  $G$  is represented by an orange rectangle containing the text  $\eta_1 \dots \eta_s$ . An arrow points from the orange rectangle to the text "random gaussians". The entire equation is enclosed in large parentheses.

```
G = randn(d, s);  
Z = exp(-sqrt(-1)*pi*X*G)/sqrt(s);
```

## OUR STARTING POINT

Want a better understanding of (and to ideally improve) the approximation guarantee:

$$\begin{array}{|c|} \hline \mathbf{Z} \\ \hline \end{array} \begin{array}{|c|} \hline \mathbf{Z}^* \\ \hline \end{array} = \begin{array}{|c|} \hline \mathbf{K} \\ \hline \end{array} + \begin{array}{|c|} \hline \begin{array}{cccc} \pm\epsilon & \pm\epsilon & \pm\epsilon & \pm\epsilon \\ \pm\epsilon & \pm\epsilon & \pm\epsilon & \pm\epsilon \\ \pm\epsilon & \pm\epsilon & \pm\epsilon & \pm\epsilon \\ \pm\epsilon & \pm\epsilon & \pm\epsilon & \pm\epsilon \end{array} \\ \hline \end{array}$$

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$$\begin{matrix} \boxed{Z} \end{matrix} \approx \begin{matrix} \boxed{Z^*} \end{matrix} = \begin{matrix} \boxed{K} \end{matrix} + \begin{matrix} \boxed{\begin{matrix} \pm\epsilon & \pm\epsilon & \pm\epsilon & \pm\epsilon \\ \pm\epsilon & \pm\epsilon & \pm\epsilon & \pm\epsilon \\ \pm\epsilon & \pm\epsilon & \pm\epsilon & \pm\epsilon \\ \pm\epsilon & \pm\epsilon & \pm\epsilon & \pm\epsilon \end{matrix}} \end{matrix}$$

- Faster but less accurate than e.g. a Nyström approximation with the same number of samples.

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- Does not give good bounds on  $\|\mathbf{Z}\mathbf{Z}^* - \mathbf{K}\|$  unless we take  $\Omega(n^2)$  samples.
- No clear implications for downstream learning tasks – e.g. does  $(\mathbf{Z}\mathbf{Z}^* + \lambda\mathbf{I})^{-1}$  approximate  $(\mathbf{K} + \lambda\mathbf{I})^{-1}$ .

In short, we want matrix-like error bounds, not entrywise bounds.

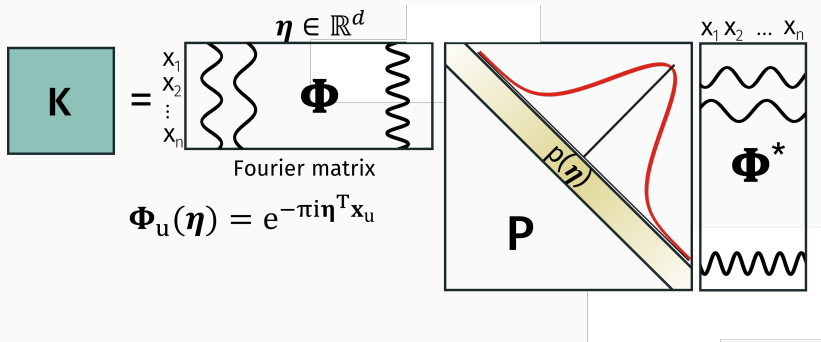
In short, we want matrix-like error bounds, not entrywise bounds.

**Key Idea:** Analyze the random Fourier features algorithm as a **matrix sampling process**.

$$k(\mathbf{x}_u - \mathbf{x}_v) = \int_{\boldsymbol{\eta} \in \mathbb{R}^d} p(\boldsymbol{\eta}) e^{-i\pi \boldsymbol{\eta}^T (\mathbf{x}_u - \mathbf{x}_v)} d\boldsymbol{\eta}$$

# HIGH LEVEL APPROACH

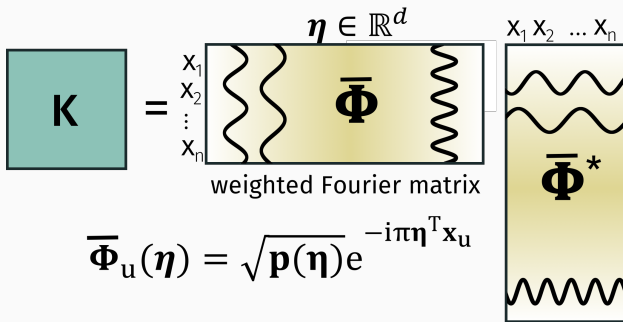
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Kernel Fourier Transform

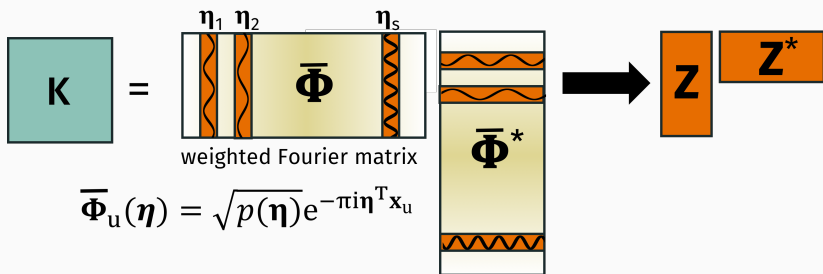
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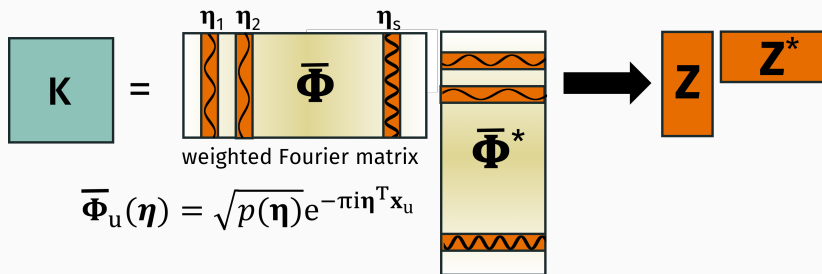


Kernel Fourier Transform

# STANDARD RFF = COLUMN NORM SAMPLING

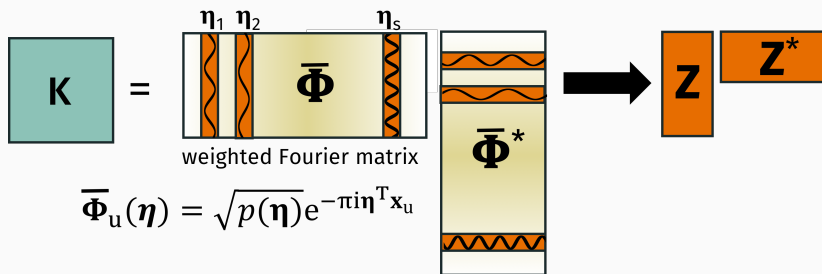


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For matrix approximation, norm based sampling probabilities are known to be suboptimal!

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We can obtain better theoretical bounds and empirical performance if we sample by column **leverage scores**.

(see work on spectral graph sparsification, matrix concentration bounds, randomized linear algebra, etc.)

**Column Norm Sampling:**  $s = O(\log n/\epsilon^2)$  samples ensure that  $(ZZ^*)_{i,j} = K_{i,j} \pm \epsilon$  for all  $i, j$  with high probability.

## COMPARISON OF APPROXIMATION GUARANTEES

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**Ridge Leverage Score Sampling:**

$$(1 - \epsilon)(ZZ^* + \lambda I) \preceq K + \lambda I \preceq (1 + \epsilon)(ZZ^* + \lambda I).$$

with  $\tilde{O}(s_\lambda/\epsilon^2)$  samples where  $s_\lambda = \text{tr}(K(K + \lambda I)^{-1})$  is the **statistical dimension**.

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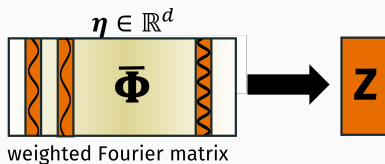
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- Since  $A \preceq B \Rightarrow B^{-1} \preceq A^{-1}$ ,  $(ZZ^* + \lambda I)^{-1} \approx (K + \lambda I)^{-1}$
- Spectral approximation also gives approximation bounds for kernel PCA, k-means, CCA, etc. (Cohen, Musco, M '16,'17)

# COMPUTING RIDGE LEVERAGE SCORES

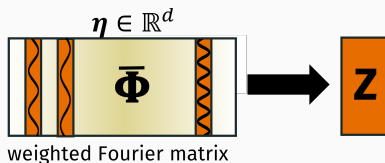
What are the ridge leverage scores?



$$\tau_{\lambda}(\eta) = \bar{\Phi}(\eta)^* (\mathbf{K} + \lambda \mathbf{I})^{-1} \bar{\Phi}(\eta).$$

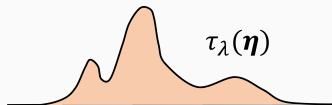
# COMPUTING RIDGE LEVERAGE SCORES

What are the ridge leverage scores?



$$\tau_{\lambda}(\eta) = \bar{\Phi}(\eta)^*(K + \lambda I)^{-1} \bar{\Phi}(\eta).$$

Expensive to invert  $K + \lambda I$ . Even if you could, not at all clear how to efficiently sample from the leverage score distribution.

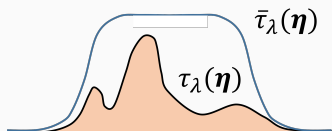


## BOUNDING THE FOURIER LEVERAGE SCORES

**Main goal:** Get a handle on the Fourier ridge leverage scores for common kernels by upper bounding them with simple distributions.

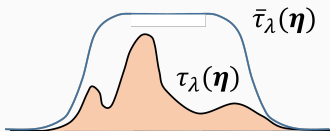
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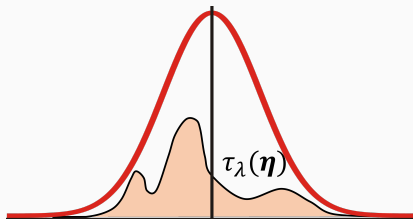
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More closely match leverage score sampling to improve random Fourier features.

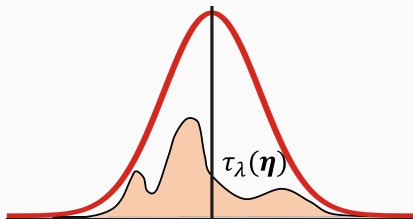
## BOUNDING THE FOURIER LEVERAGE SCORES

**First observation:** Scaled by  $\frac{n}{\lambda}$ , the standard Rahimi Recht sampling distribution upper bounds the  $\lambda$  ridge leverage scores.



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**Basic Result:** Sampling  $O(\frac{n}{\lambda} \cdot \frac{1}{\epsilon^2})$  frequencies with RFF gives spectral guarantees.

## ALTERNATIVE CHARACTERIZATION OF LEVERAGE SCORES

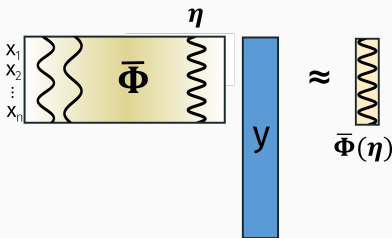
Ridge leverage score solves:

$$\tau_{\lambda}(\boldsymbol{\eta}) = \min_{\mathbf{y}} \left[ \frac{1}{\lambda} \|\bar{\boldsymbol{\Phi}}\mathbf{y} - \bar{\boldsymbol{\Phi}}(\boldsymbol{\eta})\|_2^2 + \|\mathbf{y}\|_2^2 \right].$$

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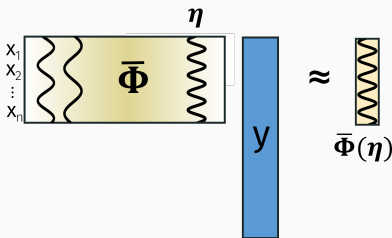
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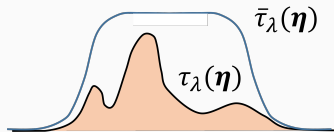


**Intuition:**  $\mathbf{y}$  reconstructs frequency  $\boldsymbol{\eta}$  from other frequencies. The easier  $\boldsymbol{\eta}$  is to reconstruct, the less important it is to sample.

## BOUNDING THE FOURIER LEVERAGE SCORES

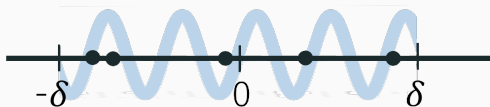
**Approach:** Obtain closed form upper bound on leverage scores by exhibiting simple candidate vector  $\mathbf{y}$  and noting that:

$$\tau_{\lambda}(\boldsymbol{\eta}) \leq \frac{1}{\lambda} \|\bar{\Phi} \mathbf{y} - \bar{\Phi}(\boldsymbol{\eta})\|_2^2 + \|\mathbf{y}\|_2^2.$$



## SIMPLIFYING ASSUMPTION

**For rest of talk:** Assume Gaussian kernel,  $n$  points in 1-dimension, bounded between  $[-\delta, \delta]$ .



## FIRST ATTEMPT

Need to bound:  $\frac{1}{\lambda} \|\bar{\Phi} \mathbf{y} - \bar{\Phi}(\boldsymbol{\eta})\|_2^2 + \|\mathbf{y}\|_2^2.$

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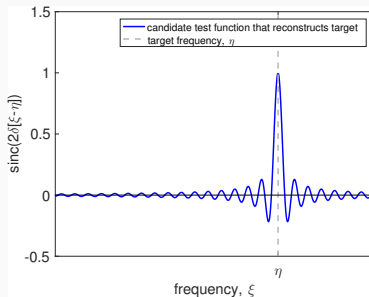
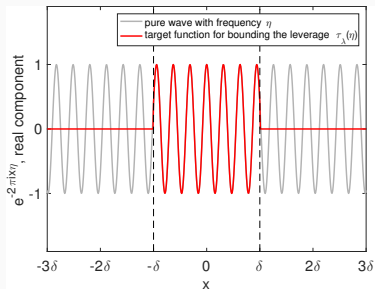
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Set  $\mathbf{y}(f)$  to be a **shifted sinc function**, weighted by  $1/\sqrt{p(f)}$ .

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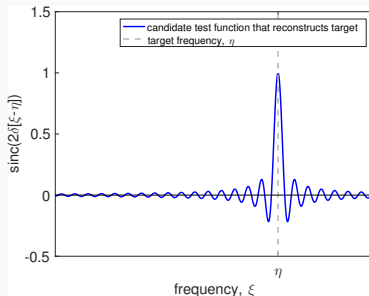
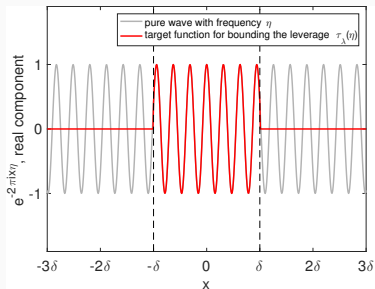
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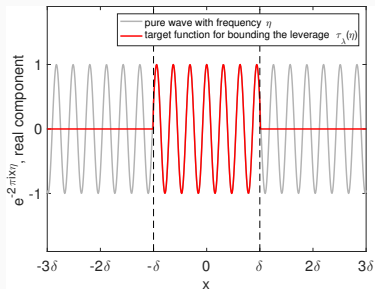


$$\frac{1}{\lambda} \|\bar{\Phi} \mathbf{y} - \bar{\Phi}(\eta)\|_2^2 = 0!$$

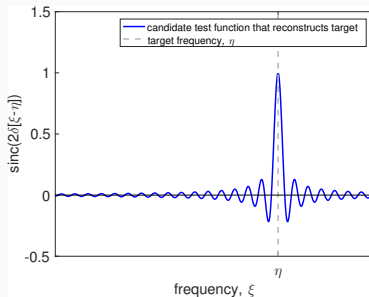
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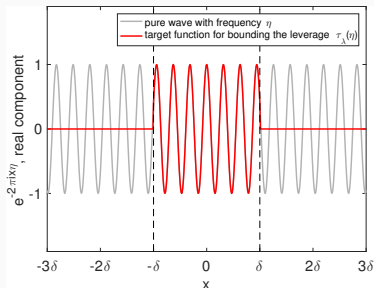


$$\|\mathbf{y}\|_2^2 = O(\delta).$$

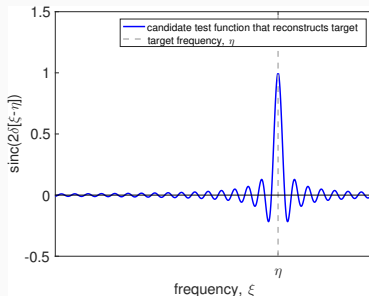
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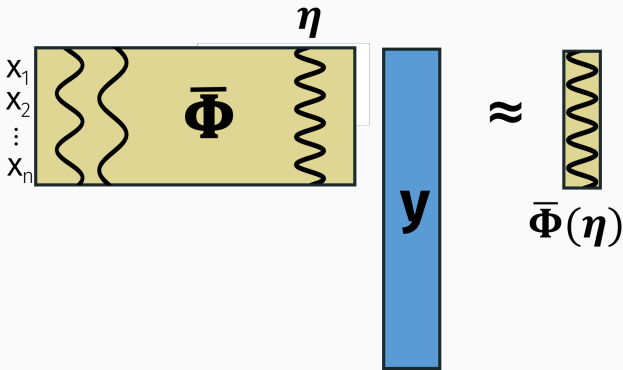


$$\|\mathbf{y}\|_2^2 = O(\delta).$$

$$\|\sqrt{p(f)} \mathbf{y}\|_2^2 = O(\delta), \quad \|\mathbf{y}\|_2^2 \text{ is much larger!}$$

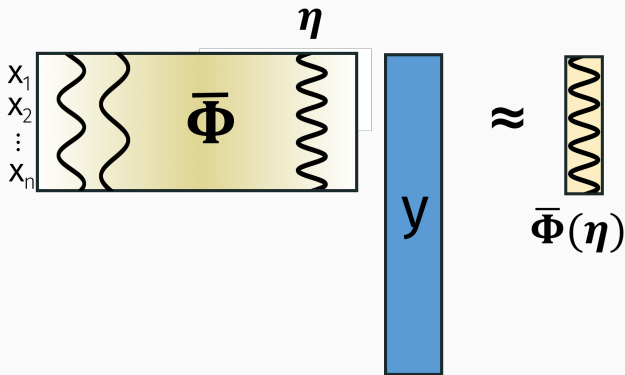
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Optimization problem is over **weighted** Fourier matrix.



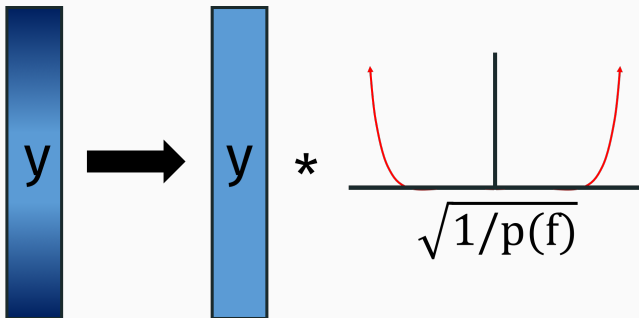
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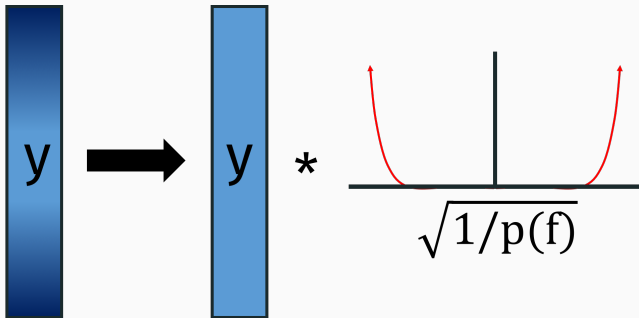
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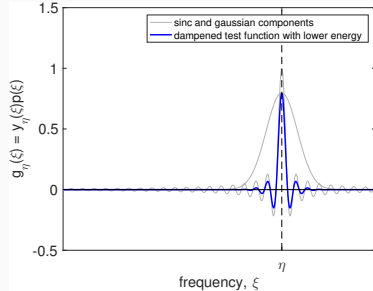
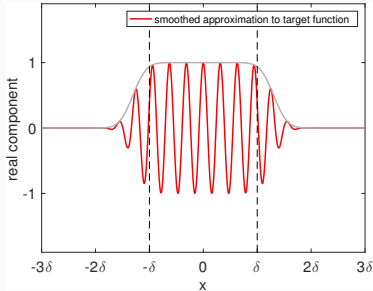
Optimization problem is over **weighted** Fourier matrix.



Sinc falls off as  $O(1/f)$ , but  $1/\sqrt{p(f)}$  grows as  $e^{O(f^2)}$ , so  $\|y\|_2^2$  explodes.

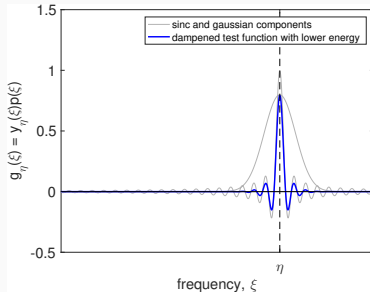
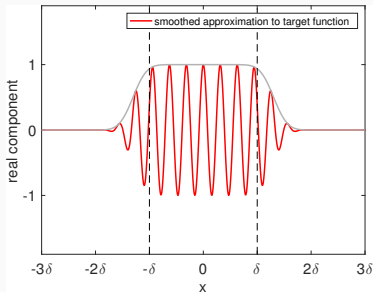
# FINAL TEST FUNCTION

**Solution:** Dampen sinc with a narrow Gaussian.



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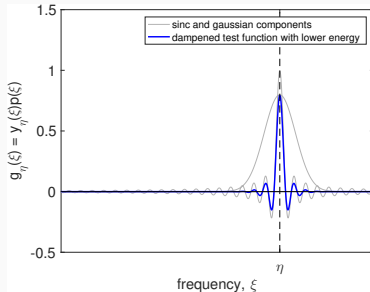
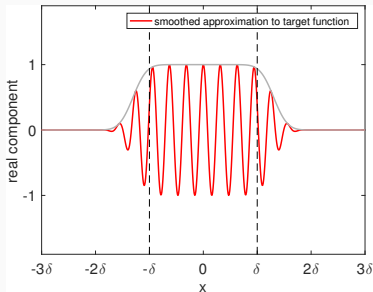
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$\frac{1}{\lambda} \|\bar{\Phi} \mathbf{y} - \bar{\Phi}(\eta)\|_2^2$  remains very small,  $\|\mathbf{y}\|_2^2 \approx O(\delta)$ .

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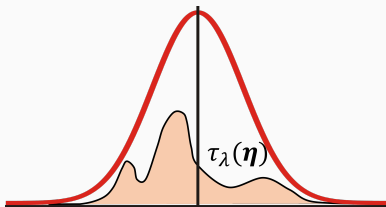
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(as long as  $\boldsymbol{\eta}$  is not too large)

## FINAL TEST FUNCTION

**Upshot:** Easy to sample from approximate leverage distribution for the Gaussian kernel with  $x_1, \dots, x_n \in [-\delta, \delta]^d$ :

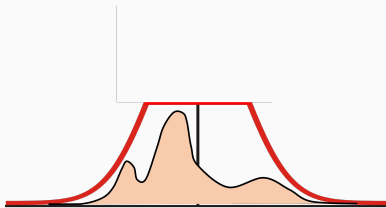
$$\bar{\tau}_\lambda(\boldsymbol{\eta}) = \begin{cases} \tilde{O}(\delta) & \text{when } \boldsymbol{\eta} \leq \sqrt{\log n / \lambda} \\ p(\boldsymbol{\eta}) = e^{-\|\boldsymbol{\eta}\|_2^2 / 2} & \text{otherwise.} \end{cases}$$



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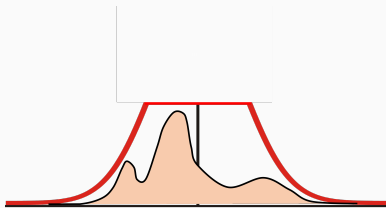
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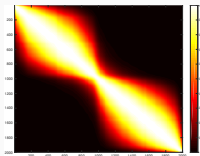
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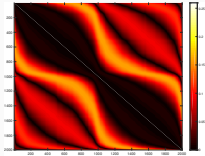
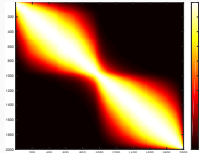
Requires  $O(\delta \sqrt{\log(n/\lambda)}) \cdot \frac{1}{\epsilon^2}$  samples for spectral guarantee.

(vs.  $O(n/\lambda) \cdot \frac{1}{\epsilon^2}$  for standard random Fourier features.)

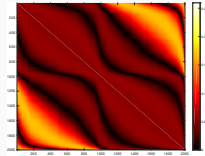
Gaussian kernel for two clusters:



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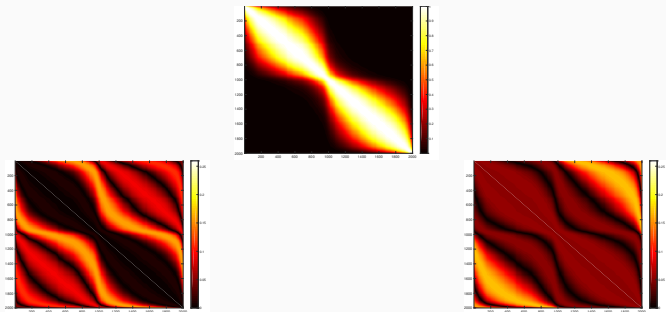


Standard RFF element error.



Modified RFF element error.

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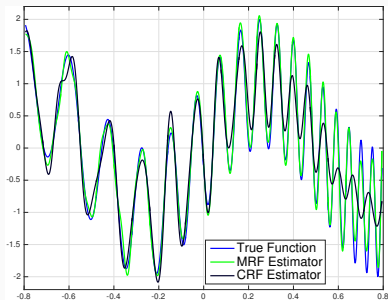
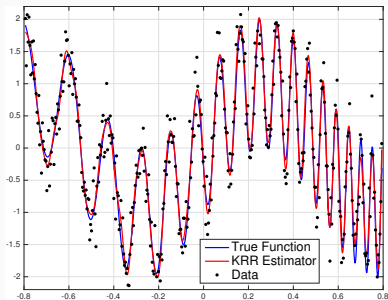
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$$(1 - \epsilon)(\mathbf{Z}\mathbf{Z}^* + \lambda\mathbf{I}) \preceq \mathbf{K} + \lambda\mathbf{I} \preceq (1 + \epsilon)(\mathbf{Z}\mathbf{Z}^* + \lambda\mathbf{I}).$$

Sampling low frequencies relatively less biases error to align with large eigenvectors.

# EXPERIMENTAL RESULTS

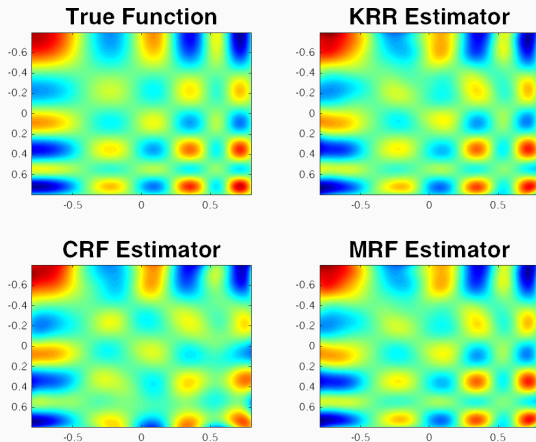
Example of approximate kernel ridge regression to interpolate a synthetic function:



CRF = classic random Fourier features ‘column norm’ sampling,  
MRF = our modified sampling distribution.

# EXPERIMENTAL RESULTS

In higher dimensions:



CRF = classic random Fourier features ‘column norm’ sampling,  
MRF = our modified sampling distribution.



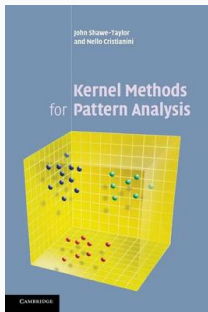
- Viewed random Fourier features methods as a **matrix sampling problem**.

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- Used optimization perspective on ridge leverage scores to certify upper bounds on these scores for bounded datasets. Reduced score computation to **Fourier approximation problem**.

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- New sampling distribution under-samples lower frequencies to obtain better low-rank approximations to the kernel matrix.

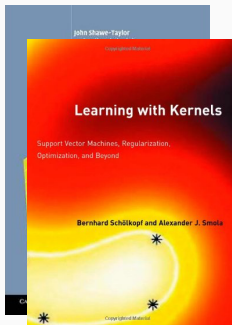
Kernel methods are widely applied at all scales.

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data analysis  
bread-and-butter

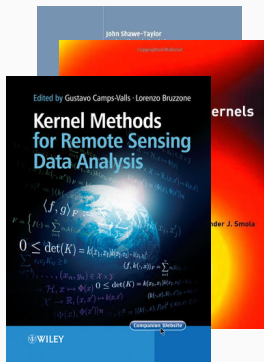
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# ALGORITHMIC CHALLENGE

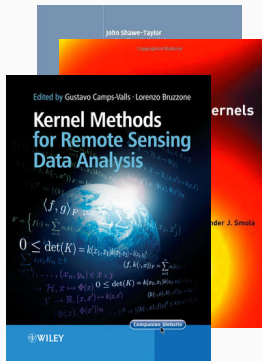
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## ALGORITHMIC CHALLENGE

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**KERNEL METHODS MATCH DEEP NEURAL NETWORKS ON TIMIT**  
*Po-Sen Huang<sup>1</sup>, Haim Avron<sup>2</sup>, Eran N. Sussan<sup>1</sup>, Huiyu Song<sup>1</sup>, Bharara Ramabhadra<sup>2</sup>*  
<sup>1</sup>Department of Electrical and Computer Engineering, University of Illinois at Urbana-Champaign, USA  
<sup>2</sup>IBM TJ. Watson Research Center, Yorktown Heights, NY, USA  
 {hsu10@illinois.edu, h.avron, eranss, vsong, bharara@watson.ibm.com}

## ABSTRACT

Despite their theoretical appeal and grounding in tractable convex optimization techniques, kernel methods are often not the first choice for large-scale machine applications due to their quadratic memory requirements and computational expense. In recent years, randomized approximate kernel methods have emerged as a large mechanism to scale-up kernel methods. Still, in practice, a large number of random features is required to achieve acceptable accuracy in predictive tasks. In this paper, we develop two algorithmic schemes to address this computational bottleneck in the context of kernel ridge regression. The first scheme is based on a novel randomized algorithm to estimate dominant components that avoids the explicit matrix-inversion of the feature space data matrix, while the second scheme gains efficiency by combining multiple weak random feature models in an ensemble learning framework. We demonstrate that these schemes outperform kernel methods to match the performance of state-of-the-art Deep Neural Networks on TMIM for speech recognition and classification tasks. We also show that our methods can be classified into two categories based on TMIM test set and word embeddings.

minimizing a regularized objective function

$$J^* = \arg\min_{J \in \mathcal{H}_0} \frac{1}{n} \sum_{i=1}^n V(\mathbf{y}_i; J(\mathbf{x}_i)) + \lambda \|J\|_k$$

$$f(x) = \sum_{i=1}^n \alpha_i h(x, a_i), \quad (1)$$

Plugging this formula back into the objective function with squared loss yields a linear system for the coefficients  $\alpha$ .

$$(N + M)\alpha = p \quad (1)$$

where  $y$  is the vector of labels and  $K$  is the  $n \times n$  Gram matrix given by  $K_{ij} = \langle \mathbf{x}_i, \mathbf{x}_j \rangle$ . It is the  $O(n^2)$  storage requirements of the Gram matrix, the  $O(n^3)$  computational expense of solving a large dense linear system during training, and the need to evaluate in  $O(n^2)$  time the sum in (1) during testing (assuming that evaluation of the kernel takes  $O(1)$  time) that make kernel methods in this form

## 1. INTRODUCTION

The recent dramatic success of Deep Neural Networks (DNNs) in speech recognition [1] highlights the statistical benefits of many low highly non-linear and non-convex models with large numbers of parameters. Efficient optimization algorithms running in distributed computing environments, their learning models propagate data through several layers of nonlinearity and learn different levels of abstraction. The composition of multiple layers of non-linear functions can approximate a rich set of naturally occurring input-output dependencies. At the same time, the combinatorial difficulty of performing exhaustive model selection in the discrete space of DNN architectures and the potential to get trapped in local minima are well recognized as valid concerns. In this paper, we revisit kernel DNNs, considered “shallow” in the DNN context. For large-scale nonparametric learning, we ask whether, despite their linear architecture and convex formulation, such models are actually scaling up kernel methods via randomized algorithms allow them to learn DNNs effectively.

To stretch the limitations in large-scale tasks, several approaches have been proposed. One of the popular directions is to approximate matrix-vector multiplication by low-rank matrices as in the Nyström method [16] and random Fourier features based methods [7, 45]. Although the Nyström method has shown better performance theoretically [15], it involves more expensive computations and larger memory requirements for large-scale tasks. Instead, our focus in this paper is on the Nyström method and its variants.

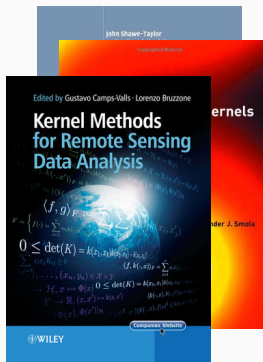
A kernel function  $k(x, y)$ ,  $x, y \in \mathbb{R}^d$  can be associated with a high dimensional inner product feature space  $\mathcal{H}$  such that  $k(x, y) = \langle G(x), G(y) \rangle$ . Our such feature space is  $\mathcal{H}$ , modified while another can be obtained through the Mercer's Theorem. For the kernel function  $k(x, y)$ , the feature space  $\mathcal{H}$  is infinite and hence it is preferred to perform computations implicitly using the kernel and its Gram matrix. However, this is at the expense of increased computational complexity with respect to the number of

kernel Hilbert Space of functions  $\mathcal{H}_K$ , with inner product  $\langle \cdot, \cdot \rangle_{\mathcal{H}_K}$  and norm  $\| \cdot \|_{\mathcal{H}_K}$ , in which model estimation can be performed by

potential for new  
state-of-the-art

# ALGORITHMIC CHALLENGE

Kernel methods are widely applied at all scales.



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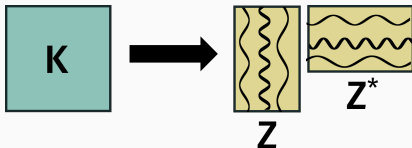
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## OPEN QUESTION

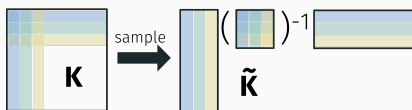
Open Question:

Kernel approximation methods you'll find in e.g.,  
`scikit-learn`:

Random Feature Methods



Nyström Methods

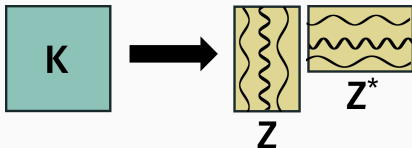


# OPEN QUESTION

## Open Question:

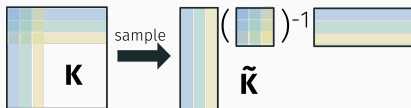
Kernel approximation methods you'll find in e.g.,  
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### Random Feature Methods



- Good accuracy, less developed theory.
- Very fast.

### Nystrom Methods



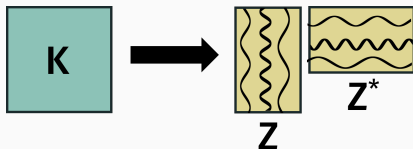
- Very good theoretical and empirical accuracy.
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# OPEN QUESTION

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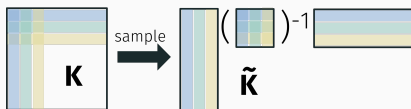
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Can either class of methods obtain optimal speed and accuracy?

Thank you!